

Methods

Assumption 1 (SUTVA)

Stable Unit Treatment Value Assumption

Potential outcomes for each unit i are unrelated to the treatment status of other individuals.

I use potential outcomes to define causal parameters as in Rubin (1974). "Estimating causal effects of treatments in randomized and nonrandomized studies". *Journal of Educational Psychology*, 66(5), 688–701.

Causal Parameters

Let Y be the variable of interest and W the treatment variable.

Y_w represents the value of the potential outcome of the interest variable based on the value w of the treatment variable W .

In the simplest case, W is a binary variable, meaning that there are two possibilities for treatment: either the unit is in the treatment group or it is not. Therefore, there are two possible values for Y_w :

* the unit belongs in the treatment group: $w = 1 \Rightarrow Y_w = Y_1$

* the unit belongs in the control group: $w = 0 \Rightarrow Y_w = Y_0$

This setup implies that the realized outcome for each particular unit depends only on the value of its corresponding treatment and does not rely on the outcome or treatment of any of the other units (SUTVA).

$$Y_{wi} = Y_i \Leftrightarrow W_i = w$$

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Assumption 2

(Unconfounded Treatment Assignment)

The potential outcomes are independent of the treatment assignment

$$(Y_1, Y_0) \perp\!\!\!\perp W$$

This implies the absence of selection bias, ensuring that the estimated Average Treatment Effect on the Treated (ATET) reflects the true causal effect. Additionally, we assume consistency in the observed outcomes.

Assumption 3

(Consistency)

The observed outcome for any unit under the treatment actually received is equal to the potential outcome for that unit under that treatment condition.

$$W_i = w \Rightarrow Y_i = Y_i(w), \quad \forall i$$

The observed outcome $Y = WY_1 + (1 - W)Y_0$ whether the unit is treated or not. However, obtaining the treatment effect for each unit $Y_{1i} - Y_{0i}$ is simply not possible because a unit cannot be both treated and untreated. This is known as the Fundamental Problem of Causal Inference. To address this, several estimation methods are employed, each relying on different identification strategies to evaluate the ATE or ATET.